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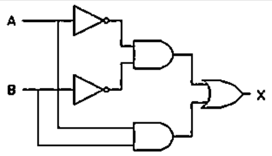
1. The following circuit generates the same ouput as?
(CUET-PG 2025) Logic Gate



(a) XOR GATE (b) XNOR GATE
(c) NAND GATE (d) NOR GATE

2. Which of the following statement is true about IEEE754 representation of +0 and -0?
(CUET-PG 2025) Floating Point (IEEE-754)
(a) have the same sign bit, but different mantissa bits.
(b) differ only in the sign bit; their exponent and mantissa bits are identical.
(c) have the same representation because they are mathematically equivalent.
(d) differ in both their expnonent and mantissa bits.

3. Perform the arithmetic addition of the two decimal numbers given in **List-I** using the signed-complement system. Match the corresponding output of **List-I** with binary number representation given in **List-II.**

List-I	**List-II**
(A) +6, +13	(I). 00000111
(B) -6, +13	(II). 00010011
(C) +6, -13	(III). 11101101
(D) -6, -13	(IV). 11111001

(CUET-PG 2025) Signed Complement
Choose the **correct** answer from the options gven below:
(a) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
(b) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
(c) (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
(d) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

4. What is the result of the following operation defined by IEEE754. (NaN = = Nan)
(CUET-PG 2025) Comparison
(a) false (b) true
(c) error (d) 1

5. Consider the following Karnaugh Map (k-map). Minimal Function generated by this Kamaugh map is

RS \ PQ	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

(CUET-PG 2025) K- map
(a) Q'S + P'Q (b) Q'S' + P'S'
(c) P'Q' + P'S' + Q'S' (d) PQ + Q'S' + P'S'

6. What should be the output of the following boolean expression after simplifying it to a minimum number of variables?
a'b' + ab + a'b
(CUET-PG 2025) Boolean Algebra
(a) a + b' (b) a' + b
(c) a' + b' (d) a + b

7. Minimum number of 2:1 Multiplexers required to design a 16:1 Multiplexer is?
(CUET-PG 2025) Multiplexar
(a)12 (b)14
(c)15 (d)16

8. Consider the following statements.
(A) Combinational logic circuits do not have memory, while sequential logic circuits have memory elements,
(B) Sequential logic circuits depend only on the current inputs, while combinational logic circuits depend on both current and past inputs.
(C) Flip-flops and latches are examples of combinational logic circuits.
(D) Multiplexer is an example of combinational logic circuit.
Choose the **correct** statement/statements from the options given below:
(CUET-PG 2025) Sequential Circuts
(a) (A) and (D) only
(b) (A), (B) and (D) only
(c) (A), (B), (C) and (D) only
(d) (B), (C) and (D) only

9. Arrange the following steps in the correct order to understand the functioning of a 3 to 8 line decoder.
(A) The encoder enable inputs is set to 1.
(B) The decoder activates one of its 8 output lines based on the input code.
(C) The input binary code is applied to the decoder.
(D) The decoder converts the 3- bit binary input into 8 possible outputs.
(CUET-PG 2025) Decodar
Choose the **correct** answer from the options given below:
(a) (A), (B), (C), (D)
(b) (A), (D), (C), (B)
(c) (A), (C), (D), (B)
(d) (A), (B), (D), (C)

10. Math List-I with List-II

List-I	List-II
J K Flip Flop inputs	**Next State**
(a) 0 0	(i). Toggle
(b) 0 1	(ii). Set
(c) 1 0	(iii). Reset
(d) 1 1	(iv). No Change

(CUET-PG 2025) Digital Electronics Flip Flop

Choose the correct answer from the options given below.

(a) (A) – (i), (B) – (ii), (C) – (iii), (D) – (iv)

(b) (A) – (iv), (B) – (iii), (C) – (ii), (D) – (i)

(c) (A) – (iv), (B) – (iii), (C) – (i), (D) – (ii)

(d) (A) – (i), (B) – (ii), (C) – (iv), (D) – (iii)

11. To provide a memory capacity of 32K X 16 how many address lines and data are required?

(CUET-PG 2025) Memory Organization

(a) address lines=14, data lines=16

(b) address lines=15, data lines=16

(c) address lines=14, data lines=4

(d) address lines=15, data lines=4

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	b	b	a	b	b	c	a	c	b
11.									
b									